is in the form: $a + bj$. The information is carried in the variables: a & b, but the proper complex number is the entire expression: $a + bj$. In polar form, the key information is contained in M & θ, but what is the full expression for the proper complex number?

The key to this is Eq. 30-9, the polar-to-rectangular conversion. If we start with the proper complex number, $a + bj$, and apply Eq. 30-9, we obtain:

EQUATION 30-10
Rectangular and polar complex numbers. The left side is the rectangular form of a complex number, while the expression on the right is the polar representation. The conversion between: M & θ and a & b, is given by Eqs. 30-8 and 30-9.

$$a + jb = M(\cos\theta + j\sin\theta)$$

The expression on the left is the proper *rectangular* description of a complex number, while the expression on the right is the proper *polar* description.

Before continuing with the next step, let's review how we arrived at this point. First, we gave the rectangular form of a complex number a graphical representation, that is, a location in a two-dimensional plane. Second, we defined the terms M & θ to be consistent with our previous experience about the relationship between polar and rectangular coordinates (Eq. 30-8 and 30-9). Third, we followed the mathematical consequences of these actions, arriving at what the correct polar form of a complex number must be, i.e., $M(\cos\theta + j\sin\theta)$. Even though this logic is straightforward, the result is difficult to see with "intuition." Unfortunately, it gets worse.

One of the most important equations in complex mathematics is **Euler's relation**, named for the clever and very prolific Swiss mathematician, Leonhard Euler (1707-1783; Euler is pronounced: "Oiler"):

EQUATION 30-11
Euler's relation. This is a key equation for using complex numbers in science and engineering.

$$e^{jx} = \cos x + j\sin x$$

If you like such things, this relation can be proven by expanding the exponential term into a Taylor series:

$$e^{jx} = \sum_{n=0}^{\infty} \frac{(jx)^n}{n!} = \left[\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}\right] + j\left[\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}\right]$$

The two bracketed terms on the right of this expression are the Taylor series for $\cos(x)$ and $\sin(x)$. Don't spend too much time on this proof; we aren't going to use it for anything.